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Akul Gopal Devali

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Cloud Platform Engineer

Portfolio: akuldevali.fyi

EDUCATION

North Carolina State University | *Raleigh, NC, USA* **Aug 2024 — May 2026**

Master of Computer Science GPA: 3.7/4

Coursework: *Algorithm Design, DevOps, Computer Networks, Cloud Computing, Software Engineering*

Vidyavardhaka College of Engineering | *Mysuru, India* **Aug 2018 — July 2022**

Bachelor of Engineering in Computer Science CGPA: 8.4/10

EXPERIENCE

DevOps Engineer Intern | *FlexGen Power Systems, Durham, NC* **June 2026 – Present**

- Refactored the production CI/CD pipeline for **HybridOS, FlexGen's energy management platform**, on AWS (CodePipeline, CodeBuild, EventBridge, Lambda), adding monorepo change detection and failure isolation to eliminate redundant builds and cut run times.
- Developed an **event driven notification service** using AWS Lambda, Slack API to report build status and failures to engineering teams.
- Configured cross-account access to **Aurora** PostgreSQL through AWS RDS Data API, using IAM role assumption and **AWS KMS** permissions to enable secure retrieval and updates of component release metadata from CI/CD pipelines.
- Migrated ~3 TB of container images across 67 repositories from Docker Hub to Amazon ECR using skopeo, with feature-flagged dual-push for a **zero disruption cutover** that preserved tags and digests.

Cloud Platform Engineer | *Liquid Rocketry Lab, Raleigh, NC* **Aug 2025 – May 2026**

- Architected and implemented a **Kubernetes-based distributed replay platform**, defining cluster topology, networking patterns and deployment workflows.
- Designed networking architecture enabling **1:1 user-to-pod communication** using Kubernetes Services and ingress routing.
- Engineered platform to scale to **20,000 concurrent replay sessions** with automated resource orchestration.

Performance Engineering Intern | *Brane Services Pvt Ltd, Bengaluru, India* **Aug 2022 - Feb 2023**

- Built internal reliability platform for 15+ microservices, combining automated **capacity benchmarking and performance analysis** to ensure stable, cost-efficient deployments.
- Achieved 25% reduction in AWS compute expenses through strategic **right-sizing of EC2 instances**, including migration of 60% workloads to spot instances and dynamic resource scheduling.
- Engineered automated load testing framework using **k6 and built Jenkins CI/CD pipelines** with **integrated performance gates**, reducing manual testing efforts by 40% while preventing regressions by 30%.

SKILLS

ML Platform & MLOps - Kubeflow, MLflow, KServe, Data Version Control (DVC), Model Registry, Model Monitoring.

Cloud Infra, Observability - AWS, Kubernetes, Grafana, Helm, Terraform, Ansible, Jenkins, GitHub Actions, Prometheus, Istio, Python, Bash

PROJECTS

MLOps Churn Prediction Platform | *Argo CD, Kubernetes, Kserve, GitHub Actions, S3* **Jan 2026 - Mar 2026**

- Designed and implemented an **end-to-end MLOps pipeline** for training, versioning and deploying a churn prediction model using **Python, scikit-learn, Kubernetes and AWS S3**.
- Built a **CI pipeline with GitHub Actions** to automate dataset generation, model training, artifact packaging and S3 model publishing.
- Implemented **GitOps deployment using Argo CD**, automatically syncing Kubernetes manifests and triggering model rollouts on Git changes.
- Deployed scalable inference endpoint using **KServe on Kubernetes**, integrating service accounts to access secrets from AWS.

Containerized Microservices Deployment using GitHub Actions | *EC2, Ansible, k6, GitFlow* **Aug 2025 - Dec 2025**

- Built a robust **CI/CD** pipeline for a containerized Node.js application, utilizing **GitHub Actions** for orchestration and **Docker Hub** for artifact management.
- Implemented **shift-left security practices** by integrating **pre-commit hooks** for static code analysis and dependency audits in local and development environments, and configured CI workflows to scan every pull request for secret leaks and vulnerabilities prior to merge.
- Developed **Ansible playbooks** to automate config management, used dynamic inventories to isolate Staging and Production environments.
- Implemented a **Performance Quality Gate using k6**, preventing performance regressions by automatically halting production **rollouts** if load tests fail in the staging environment.

ACHIEVEMENTS AND EXTRA-CURRICULAR

- Awarded AWS All Builders Welcome Grant to attend AWS re:Invent Las Vegas 2025 for contribution towards Cloud and DevOps learning.
- 2x AWS Certified Solutions Architect Associate and Google Cloud Platform specialization in Architecting with Google Compute Engine.